# A ten-point deficit progression in the mod-107 signed-divisor support

Erdős–Straus Workbench
community lead credited to `u/CommonCareful3149`

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**Abstract**

For the record integer $p = 8{,}803{,}369$ and residual shell $R = 107$, the even-factor half-log support has a ten-point complement in $\mathbb{Z}/53\mathbb{Z}$. We verify that this complement is the arithmetic progression $D = \{23 + 42j : 0 \le j \le 9\}$ and that $|D - D| = 19$. Since the gap between the two odd-sheet translates is not in $D - D$, the translates saturate the full quadratic-nonresidue coset modulo 107. This is one bounded support calculation, not a general shell theorem, an endpoint proof, or a priority claim.

## 1 Provenance and scope

A public post by Reddit user u/CommonCareful3149 reports an investigation of the mod-107 Erdős–Straus shell and announces a short technical note. This workbench credits the progression and difference-set observation to that community lead and reconstructs it independently below. The unpublished note and private correspondence are not reproduced.

The exact progression language was not found in the routed local project corpus; the factor logarithms, parity sheets, target fibres, and missing exponent classes were already present there. This local search result does not establish global novelty. The immutable historical baseline is Erdős–Straus Project Archive, Zenodo record 21845035.

## 2 Coordinates

Set

$$p = 8{,}803{,}369, \qquad R = 107, \qquad a = \frac{p+R}{4} = 2{,}200{,}869 = 3^2 11^2 43\,47.$$

The element 2 has order 106 modulo 107. Hence

$$\lambda : (\mathbb{F}_{107}^{\times}, \cdot) \longrightarrow C_{106} := \mathbb{Z}/106\mathbb{Z}, \qquad \lambda(u) = \log_2(u),$$

is an isomorphism, with inverse $e \mapsto 2^e$. Direct computation gives

$$\lambda(3) = 70, \qquad \lambda(11) = 22, \qquad \lambda(43) = 59, \qquad \lambda(47) = 66. \tag{1}$$

Thus 43 is the unique factor with odd logarithm.

The quadratic residues correspond under $\lambda$ to the even subgroup $2C_{106}$. Define the exact half-log isomorphism

$$h : 2C_{106} \longrightarrow C_{53} := \mathbb{Z}/53\mathbb{Z}, \qquad h(2x) = x,$$

with inverse $x \mapsto 2x$. Removing the factor 43, the centered even-factor support maps to

$$A = \{35r + 11s + 33t : -2 \le r, s \le 2,\ -1 \le t \le 1\} \subseteq C_{53}. \tag{2}$$

The map $h$ has trivial kernel and singleton fibres; this change of coordinate loses no information.

## 3 The deficit and its difference set

**Theorem 1.** *The set $A$ has 43 elements. Its complement is*

$$\begin{aligned} D := C_{53} \setminus A &= \{1, 10, 12, 21, 23, 30, 32, 41, 43, 52\} \\ &= \{23 + 42j \pmod{53} : 0 \le j \le 9\}. \end{aligned} \tag{3}$$

*Moreover,*

$$D - D = \{42k \pmod{53} : -9 \le k \le 9\}, \qquad |D - D| = 19 = 2|D| - 1. \tag{4}$$

*Proof.* Exact reduction of the $5 \cdot 5 \cdot 3 = 75$ triples in (2) gives $|A| = 43$ and the first list in (3). Starting at 23 and repeatedly adding 42 gives

$$23, 12, 1, 43, 32, 21, 10, 52, 41, 30,$$

which proves the progression identity.

Every difference is $42(j - i)$ with $-9 \le j - i \le 9$, and each integer in that interval occurs. Because 42 is invertible modulo 53, a collision would imply $53 \mid (k - \ell)$. But $|k - \ell| \le 18 < 53$, so $k = \ell$. Therefore all 19 displayed differences are distinct. □

## 4 Saturation of the nonresidue sheet

The odd exponent coset is a torsor, not a subgroup. We therefore use the affine bijection

$$\psi : 1 + 2C_{106} \longrightarrow C_{53}, \qquad \psi(2x + 1) = x, \qquad \psi^{-1}(x) = 2x + 1.$$

Since

$$59 = 2 \cdot 29 + 1, \qquad -59 \equiv 47 = 2 \cdot 23 + 1 \pmod{106},$$

the two nonzero centered choices for the 43 exponent give supports $A + 29$ and $A + 23$. Their complements are $D + 29$ and $D + 23$.

**Theorem 2.** *The two translated deficits are disjoint, and hence*

$$(A + 29) \cup (A + 23) = C_{53}.$$

*Under $\psi^{-1}$ and exponentiation by 2, the two centered 43-factor choices saturate the entire quadratic-nonresidue coset in $\mathbb{F}_{107}^{\times}$.*

*Proof.* An intersection would give $d_1 + 29 = d_2 + 23$, so $6 = d_2 - d_1 \in D - D$. But

$$42^{-1} = 24 \pmod{53}, \qquad 24 \cdot 6 = 144 \equiv 38 \equiv -15 \pmod{53}.$$

The integer $-15$ is not in $[-9, 9]$, so (4) implies $6 \notin D - D$, a contradiction. Taking complements proves the union identity. Both subsequent coordinate maps are bijections. □

In the alternative Legendre-sheet coordinate $v \mapsto h(\lambda(-v))$, the same supports are $A+3$ and $A-3$. The two affine presentations differ by a translation and have the same decisive gap 6.

**Corollary 3.** *The full centered exponent support*

$$B = \{70r + 22s + 59u + 66t : -2 \le r, s \le 2,\ -1 \le u, t \le 1\} \subseteq C_{106}$$

*has* 43 *even and* 53 *odd classes. Thus* $|B| = 96$ *and*

$$C_{106} \setminus B = 2D = \{2, 20, 24, 42, 46, 60, 64, 82, 86, 104\}.$$

*The target exponent* $53 = \lambda(-1)$ *has exactly the two coefficient fibres*

$$(-2, 0, -1, -1), \qquad (2, 0, 1, 1),$$

*whereas the exterior target exponent* $60 = \lambda(-p^{-1})$ *has none.*

*Proof.* Only $59u$ controls parity. The case $u = 0$ gives the 43-point even support; $u = \pm 1$ gives the saturated 53-point odd support. The complement is the inverse half-log image $2D$. Direct enumeration of the finite coefficient box gives the two stated fibres and the empty exterior fibre. □

# 5 Reproducibility and nonclaims

The standard-library checker

`certificates/verify_r107_deficit_progression.py`

reconstructs the primitive-generator table, both support coordinates, $D$, $D - D$, both translate presentations, residue-level nonresidue saturation, the full missing set, and both target fibres. Ordinary and optimized Python runs must have byte-identical output; the file contains no removable `assert` nodes.

The Lean file

`formal/lean/R107DeficitProgression.lean`

independently checks the finite-set equalities in ZMod(53) and ZMod(106) and the factor logarithms in ZMod(107). Formal acceptance and correspondence with this informal argument are separate review obligations.

**Remark 4.** *Support saturation does not by itself retain representation multiplicities; the fibre enumeration is a separate calculation. Nothing here proves a family of shells, a density theorem, or the Erdős–Straus conjecture. The workbench makes no claim that the progression observation is new.*